⚡ PROTOCOL PITCH DECK

BOTCHAIN TESTNET • CHAIN ID: 968

LIGHTBID

Every Bid Counts. Every Auction Ends Fair. Zero MEV.

Next-generation decentralized English auction engine for NFTs and digital assets on Botchain, featuring dynamic anti-snipe clock extensions, pull-payment escrow, and native BOT settlement.

10 MIN	100%	3 ASSETS	~3 SEC
ANTI-SNIPE EXTENSION	PULL-PAYMENT ESCROW	ERC-721 / 1155 / 20	BOTCHAIN FINALITY

Contract: `0x39FfB249F207a2583Be272d2f55AcebE0d09b2b8` • RPC: `https://rpc.bohr.life` • Tokens: **BOT / tUSD / LNFT**

The Problem: Legacy Web3 Auctions are Broken

Traditional on-chain auctions suffer from predatory MEV bot front-running, contract-level DoS attack vectors, and high friction that destroy true price discovery.

🤖 Last-Second Sniping

Conventional auctions end at a static block timestamp. Automated sniper bots monitor the mempool and broadcast higher bids with exorbitant priority gas in the final 500ms, completely boxing out human bidders.

- Zero time for legitimate bids to respond
- Suppresses final clearing prices for creators
- Enriches MEV searchers at users' expense

🛑 Push-Payment DoS

Standard marketplace contracts automatically "push" refunded funds back to the previous bidder inside the `bid()` transaction. If a malicious bidder's contract reverts, the entire auction is frozen forever.

- Reverting `receive()` breaks bidding loop
- Attacker locks in low bid as winning bid
- Reentrancy and state mutation risks

🔄 Wrapping & High Fees

Platforms on Ethereum and L2s force users to wrap native currency into WETH or ERC-20 proxies, generating extra approval transactions, gas burn, and friction for retail collectors and gamers.

- Multi-step approval & wrap workflows
- High gas fees ($5–$30 per transaction on L1)
- Unnecessary cognitive overhead for users

🗂️ Fragile Metadata

Many Web3 auction platforms rely on centralized Web2 AWS S3 buckets or temporary API gateways for media and metadata storage, resulting in dead links, 404 errors, and disappearing artwork.

- Centralized single point of failure
- High risk of NFT media "link rot"
- No integrated creator minting pipeline

The Solution: The Lightbid Engine

A mathematically fair, zero-MEV English auction protocol engineered from the ground up for Botchain’s sub-second execution environment.

⏱️ Dynamic Anti-Snipe

Whenever a valid bid is placed within 10 minutes of scheduled auction completion (`EXTENSION_WINDOW`), the clock automatically extends by an additional 10 minutes.

- Guarantees human bidders fair reaction time
- Simulates natural competitive live auction room
- Maximizes authentic asset value for creators

🛡️ Pull-Payment Escrow

Outbid funds are isolated into an internal ledger (`pendingReturns[bidder]`). Bidders withdraw accumulated funds at any time with one click, completely immune to external revert exploits.

- Zero possibility of auction denial-of-service
- Reentrancy-guarded Checks-Effects-Interactions
- One-click aggregated withdrawal across all bids

⚡ Native BOT Settlement

Bidding and settlement occur directly in Botchain native **BOT**. Zero wrapped tokens. Zero intermediate token approvals. Sub-cent gas fees and instant ~3-second block finality.

- Immediate bidding directly from native wallet
- Sub-cent transaction gas execution
- Seamless user onboarding without WETH hurdles

📦 Multi-Asset & IPFS

A single marketplace contract supporting ERC-721 1-of-1s, ERC-1155 editions, and ERC-20 token bundles, natively connected to Pinata Cloud IPFS for decentralized immutable storage.

- Universal auction standard for all digital items
- In-app NFT minting directly to Pinata IPFS
- OpenSea-compliant dynamic metadata JSON

Architecture & State Machine

Rigorous smart contract lifecycle ensuring mathematical determinism, gas efficiency, and multi-asset isolation.



⏱️ Anti-Snipe Implementation (Solidity)

```
uint256 public constant EXTENSION_WINDOW = 10 minutes;
uint256 public constant EXTENSION_DURATION = 10 minutes;

if (auction.endTime - block.timestamp <= EXTENSION_WINDOW) {
    auction.endTime += EXTENSION_DURATION;
    emit AuctionExtended(auctionId, auction.endTime);
}
```

🛡️ Safe Pull-Payment Ledger (Reentrancy-Proof)

```
// Checks-Effects-Interactions Pattern
uint256 refund = pendingReturns[msg.sender];
if (refund == 0) revert NoFundsToWithdraw();
pendingReturns[msg.sender] = 0; // State updated first
(bool sent, ) = msg.sender.call{value: refund}("");
if (!sent) revert RefundFailed();
```

Multi-Asset Standards & Pinata Creator Studio

Universal asset compatibility paired with a full Web3 minting suite powered by Pinata Cloud IPFS.

💎 ERC-721 Digital Art

Full support for non-fungible 1-of-1 digital masterpieces, photography, music, and virtual world parcels.

- Integrates `safeTransferFrom & IERC721Receiver`
- Direct metadata extraction and high-res preview
- Test token deployed: **LNFT (0xD8c9...B63b)**
- Royalties and creator ownership preserved

🎮 ERC-1155 Editions & Gaming

Engineered for gaming economies, event passes, tradeable weapons, and limited-run multi-edition drops.

- Semi-fungible batch transfer support
- Implements `IERC1155Receiver` hooks
- Fractional auctioning of edition bundles
- Sub-cent gas efficiency for in-game item trade

💰 ERC-20 Token Bundles

Auction off allocations of project tokens, liquidity provider (LP) shares, or OTC token blocks.

- Built with OpenZeppelin `SafeERC20`
- Admin token allowlist protection for security
- Test token deployed: **tUSD (0x249D...0d82d)**
- Ideal for community OTC and treasury diversification

🌐 Integrated Pinata Cloud IPFS Pipeline

Creators drag-and-drop artwork directly inside the dApp. The engine uploads the file to Pinata IPFS, builds an OpenSea-compliant JSON manifest, pins the metadata, and prompts an on-chain mint transaction in under 10 seconds.

Pinata JWT Authenticated | CIDv1 Pinned | 100% Decentralized

Live On-Chain Verification on Botchain

Not a prototype or mockup. Lightbid is deployed, fully configured, and verified live on Botchain Testnet (Chain ID: 968).

26 / 26
FOUNDRY TESTS PASSED
Unit, fuzzing & reentrancy suites

3 CONTRACTS
DEPLOYED & VERIFIED
Marketplace + ERC20 + ERC721

24 TXS
MULTI-WALLET ACTIVITY
8 transactions across 3 test wallets

100%
TX SUCCESS RATE
Zero reverts or failed calls

CONTRACT NAME	CONTRACT ADDRESS	STATUS	ON-CHAIN VERIFICATION
⚡ Lightbid Marketplace	0x39FfB249F207a2583Be272d2f55AcebE0d09b2b8	● Live & Deployed	View on Bohr Scan ↗
💵 TestERC20 (tUSD Faucet)	0x249D23b4c2e58cf40eB0273e6afCE8d23170d82d	● Faucet Active	View on Bohr Scan ↗
🖼️ TestERC721 (LNFT Mintable)	0xD8c92E8e8ECe7F00C947eB2C12Cdb2A110f9B63b	● Mint Studio Ready	View on Bohr Scan ↗
👥 Multi-Wallet Load Testing	Wallets: 0x9a06...6Ef9 | 0x6738...3794 | 0x4276...98b9	● 24 Txs Executed	Confirmed in Blocks 28721904–28721967

Production-Grade Frontend & User Experience

A glassmorphic Web3 application engineered for maximum clarity, real-time reactive feedback, and zero cognitive friction.

⏱️ Live Countdown UI

Precision high-frequency countdown timers display remaining auction duration with automatic visual alerts when anti-snipe extensions trigger.

- Dynamic color-coded state pills
- Real-time block synchronization
- Anti-snipe notification banners

📊 Smart Bidding Engine

Automatic minimum next bid calculator based on auction basis points (BPS), preventing sub-increment transaction reverts before submission.

- Pre-validates user wallet balance
- 1-click increment presets (+5%, +10%)
- Instant gas estimation on Botchain

🔔 Sticky Refund Drawer

Persistent navigation bar indicator alerts bidders whenever they have accumulated outbid refunds waiting in the escrow ledger.

- Single-click claim across all bids
- Aggregated escrow balance tracker
- Zero forgotten or trapped funds

💧 In-App Token Faucet

Dedicated testnet token faucet dispensing 1,000 tUSD with cooldown protection and automatic asset addition to MetaMask.

- 1,000 tUSD per 24 hours
- One-click MetaMask token registration
- Enables immediate bundle testing

⚡ Built for rapid scale: **Lightweight client bundle** (0 linter errors, TypeScript strict, sub-1s page transitions).

Vite 5 Production Build Ready

Market Opportunity & Protocol Economics

Capturing high-velocity digital asset auction volume through a non-extractive, volume-aligned fee model.

$102B

TAM: GLOBAL DIGITAL AUCTIONS BY 2030

Encompasses digital art, gaming assets, metaverse land, and tokenized real-world assets (RWAs).

$18B

SAM: HIGH-THROUGHPUT EVM VOLUME

Annual transaction volume migrating to next-generation low-fee, high-speed L1/L2 networks like Botchain.

$50M

SOM: TARGET YEAR 1 GMV

Target initial market capture on Botchain ecosystem across gaming, launchpads, and creator drops.

1. Protocol Settlement Fee (2.5%)

A standard 250 BPS (2.5%) fee deducted from the final sale proceeds upon successful settlement. Capped by contract at 10% to protect creators.

- At $20M GMV = $500,000 protocol revenue
- At $50M GMV = $1,250,000 protocol revenue

2. Curated Launchpad Drops

Premium homepage placement, marketing syndication, and verified badge auctions for flagship game releases and top digital artists.

- Fixed curation & listing tier fees
- Co-marketing revenue share

3. Enterprise White-Label SDK

Turnkey auction widgets and smart contract factory deployments for external Web3 games and DAO treasury liquidation sales.

- SDK licensing & API access
- Custom secondary royalty split contracts

Competitive Advantage & Feature Matrix

How Lightbid fundamentally outperforms incumbent marketplaces across security, fairness, and asset flexibility.

FEATURE DIMENSION	OPENSEA / ZORA	FOUNDATION	⚡ LIGHTBID (BOTCHAIN)
Anti-Snipe Protection	15 min (ERC-721 only)	15 min (Fixed)	**Dynamic 10 min (ERC-721, 1155, 20)**
Bid Refund Architecture	Push Payment (Revert DoS Risk)	Push Payment (Revert DoS Risk)	**Safe Pull-Payment Escrow (Zero DoS)**
Supported Asset Standards	ERC-721 primarily	ERC-721 only	**Universal: ERC-721, ERC-1155, ERC-20**
Currency Settlement	Wrapped Tokens (WETH)	Wrapped Tokens (WETH)	**Native BOT (Zero Wrapping Friction)**
Average Gas Cost per Bid	$5.00 – $25.00 (L1)	$5.00 – $25.00 (L1)	**Sub-Cent (< $0.001 on Botchain)**
Settlement Finality	12 – 15 Seconds	12 – 15 Seconds	**~3 Seconds Instant Finality**
Creator IPFS Mint Studio	Separate Studio Tool	No In-App Studio	**Built-In Pinata Cloud IPFS Pipeline**
Emergency Safeguards	Centralized Operator Registry	Basic Pause	**Pausable + Ownable2Step + ReentrancyGuard**

Strategic Roadmap & Milestone Execution

Disciplined phased execution scaling from robust testnet foundation to enterprise cross-chain auction infrastructure.

PHASE 1 • Q4 2026

Foundation (Complete)

Core smart contracts deployed, verified, and load-tested live.

- English auction with anti-snipe
- Pull-payment escrow system
- 26/26 Foundry tests passing
- Botchain Testnet deployment
- Pinata IPFS creator pipeline
- 24 multi-wallet live transactions

PHASE 2 • Q1 2027

Format Expansion

Multi-format auction mechanics and mainnet launch preparation.

- Descending Dutch auction engine
- Sealed-bid blind auction module
- Automated EIP-2981 royalty splits
- Tier-1 smart contract audit
- Botchain Mainnet deployment
- First 10 flagship creator drops

PHASE 3 • Q2 2027

Developer SDK & Mobile

Extending auction rails to third-party dApps and mobile devices.

- Lightbid npm SDK for games
- Real-time WebSocket orderbook
- Automated bidding bot framework
- Mobile PWA with push alerts
- DAO treasury auction modules
- Creator royalty management UI

PHASE 4 • Q3 2027

Institutional & RWA

Cross-chain liquidity and institutional tokenized assets.

- RWA compliance & legal framework
- Cross-chain bidding (CCIP)
- Multi-sig institutional custody
- Enterprise phygital luxury drops
- Governance token & DAO launch
- Global auction syndication

INVESTMENT OFFERING

$1,250,000 Seed Round

Accelerating engineering, smart contract formal verification, and creator onboarding to establish Lightbid as the premier decentralized auction layer on Botchain.

Capital Allocation:

⚙️ Core Protocol & SDK Engineering	45% ($562.5k)
🛡️ Smart Contract Audits & Security	25% ($312.5k)
🎨 Creator Grants & Ecosystem Liquidity	20% ($250.0k)
⚖️ Legal, Operations & Compliance	10% ($125.0k)

ON-CHAIN PROTOCOL CREDENTIALS

Verified Live on Botchain Testnet

```
// Core Contract Address
0x39FfB249F207a2583Be272d2f55AcebE0d09b2b8

// Network Infrastructure
Chain ID: 968 | RPC: https://rpc.bohr.life

// Block Explorer
https://scan.bohr.life/address/0x39Ff...b2b8 ↗
```

Ready to Partner with Lightbid:

We welcome early angel investors, creator collectives, gaming studios, and Botchain ecosystem builders to participate in the future of fair Web3 auctions.